Melt-pool uniformity optimization of a serpentine raster: exact per-segment Newton control with OTI derivatives

Notes on `melt_pool_demos/raster` (3DThesis + `cpp_oti_lib`)

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Abstract

These notes document how the raster uniformity optimization works, end to end: the semi-analytical forward model (3DThesis), how exact melt-pool sensitivities are obtained from a single simulation using order-truncated imaginary (OTI) numbers, how field derivatives become derivatives of the pool *dimensions* through the implicit function theorem, and the damped per-segment Newton controller built on top. The demo holds the melt-pool width and depth of a 1×1 mm serpentine raster to a fixed target while heat accumulates, reducing the width/depth coefficient of variation from 9.7%/7.1% to 0.71%/0.59% in 49 simulations ($\approx$65 s wall clock, single process). Section 8 extends the study to a 10×10 mm square at the published EBM-scale parameters of Stump and Plotkowski's triangle experiment (150 W absorbed, 3 m/s, $\sigma_{xy} = 200\,\mu$m, IN718 at 1273 K), where the pool is millimetres long and merges across hatch lines. Judged on whole-build statistics — the beam-on fluctuation of the instantaneous pool dimensions — the controller cuts the pool-volume standard deviation $\approx 4.5\times$ (from 0.077 ± 0.024 to $0.020 \pm 0.005\,\text{mm}^3$ at zero dwell) while pinning the mean to the requested single-track setpoint; removing the turnaround dwell trades a modestly tighter build (fusion-depth σ 4.4 vs. 6.7 μm) against 26% of build time.

1 Problem statement

A laser scans a 1×1 mm square in a serpentine (boustrophedon) raster: 7 lines at speed $V = 0.7$ m/s with hatch spacing $H = 0.15$ mm, with a short beam-off dwell at every turnaround. As the part heats up, a fixed-power beam produces a melt pool that balloons — both line to line (width grows $97 \to 135\,\mu$m over the square) and within each line (each track starts small on freshly solidified ground and swells toward its end).

The goal is *geometric uniformity*: hold the pool's half-width w and depth d at a fixed target (w^*, d^*) for every control segment of the path. Two controls are available per 0.5 mm segment i:

$$
\begin{aligned}
P_{\text{mod},i} &\quad \text{— power multiplier; segment power } Q_i = P_{\text{mod},i}\, Q_{\text{base}}, \\
\sigma_i &\quad \text{— lateral beam width (the Gaussian } \sigma_x = \sigma_y \text{ of the source).}
\end{aligned}
$$

Power scales how much material melts; beam width *reshapes* the pool at fixed power (wider beam = same energy over a larger footprint = broader, shallower pool). Both knobs are needed: power alone cannot hold a fixed (w^*, d^*) pair once background preheat exists, because it only scales the pool while the target also constrains its aspect ratio.

2 Forward model: semi-analytical conduction (3DThesis)

The thermal model is the adaptive-quadrature conduction solver of Stump and Plotkowski [1], implemented in 3DThesis. Assuming constant properties and pure conduction, energy conservation is

$$\rho c \frac{\partial T}{\partial t} = k\nabla^2 T + \dot{Q}, \tag{1}$$

which, for a volumetric Gaussian source moving along an arbitrary path on a semi-infinite domain, has the transient integral solution (Nguyen et al. [2])

$$T(\mathbf{x}_p, t) - T_0 = \frac{2\eta Q}{\rho c\,(\pi/3)^{3/2}} \int_0^t \frac{1}{\sqrt{\phi_x\phi_y\phi_z}} \exp\!\left(-\frac{3x(t')^2}{\phi_x} - \frac{3y(t')^2}{\phi_y} - \frac{3z(t')^2}{\phi_z}\right) \mathrm{d}t', \tag{2}$$

with $\phi_i = 12\alpha\,(t - t') + \sigma_i^2$ and $\mathbf{x}(t') = \mathbf{x}_p - \mathbf{x}_b(t')$ the distance from the evaluation point to the (piecewise-linear) beam position. Nondimensionalizing on the lateral beam width σ_{xy} [1],

$$s = \frac{\alpha}{\sigma_{xy}^2} t, \qquad X = \frac{x}{\sigma_{xy}}, \quad Y = \frac{y}{\sigma_{xy}}, \quad Z = \frac{z}{\sigma_{xy}}, \qquad \delta = \frac{\sigma_z}{\sigma_{xy}}, \tag{3}$$

puts the integrand into the universal form

$$f(s'') = \frac{1}{(12s''+1)\sqrt{12s''+\delta}} \exp\!\left(-\frac{3\left(X^2+Y^2\right)}{12s''+1} - \frac{3Z^2}{12s''+\delta}\right), \tag{4}$$

where $s'' = s - s'$ is the conduction time available for heat deposited at s' to diffuse. 3DThesis evaluates (2) by Gauss–Legendre quadrature over sub-segments of the path history, with the two adaptive rules of [1]: the segment size doubles as $\Delta s'' = 2^{\lfloor \log_2(s''+1)\rfloor}$ while the quadrature order halves from 16 down to 2 (diffusion makes f increasingly linear), and for a moving source the node spacing is capped by

$$\Delta s'' \;=\; \frac{\sqrt{(12s''_n+1)\,\ln\sqrt{2}}}{V^*}, \qquad V^* = \frac{\sigma_{xy}\,\bar{v}}{\alpha}, \tag{5}$$

so that neighboring discrete sources overlap into a smooth field. The end result: the temperature at any point is a *finite weighted sum of Gaussian contributions from quadrature nodes along the path history*,

$$T(\mathbf{x}_p) = T_0 + \sum_{n\in\text{nodes}} c_n \exp\!\left(-\frac{3\,(x_p - x_{b,n})^2}{\phi_{x,n}} - \frac{3\,(y_p - y_{b,n})^2}{\phi_{y,n}} - \frac{3\,(z_p - z_{b,n})^2}{\phi_{z,n}}\right), \tag{6}$$

where for a node deposited at time t'_n carrying power Q_n,

$$c_n = \frac{2\eta\,Q_n}{\rho c\,(\pi/3)^{3/2}} \frac{w_n}{\sqrt{\phi_{x,n}\,\phi_{y,n}\,\phi_{z,n}}}, \qquad \phi_{i,n} = 12\,\alpha\,(t - t'_n) + \sigma_i^2, \tag{7}$$

w_n is the Gauss–Legendre weight of the node (times its sub-interval length), $\alpha = k/\rho c$, and $t - t'_n$ its conduction time. This is (4) restored to dimensional form and summed by quadrature: the Gaussian widths ϕ_i broaden with conduction time (diffusion), and c_n carries the source strength together with the $1/\sqrt{\cdot}$ normalization. The compact notation $\|\mathbf{x}_p - \mathbf{x}_{b,n}\|^2_{\Phi_n} \equiv \sum_i (x_{p,i} -$

$x_{b,n,i})^2/\phi_{i,n}$ abbreviates the exponent below. The per-segment controls enter (6) transparently: P_{mod} scales Q_n for the nodes of its segment, σ enters $\phi_{x,n}, \phi_{y,n}$ (3DThesis's path columns `Pmod`, `Wmod`), and the scan speed v enters only through the node deposition times t'_n (Section 3.1).

Every optimization "simulation" below is one such solve on a small local snapshot domain (10 μm resolution, 0.5 mm buffer) placed at the end of the truncated path, i.e. around the current pool.

3 Exact derivatives in one pass: OTI numbers

The optimizer needs derivatives of the pool dimensions with respect to the controls. Rather than finite-differencing (3 simulations per Jacobian, step-size tuning, subtractive cancellation), the OTI build of 3DThesis computes them *analytically in the same run.*

The scalar type of the entire solver is swapped (`Real = oti::otinum<6,1,double>` from `cpp_oti_lib`): every `Real` carries a value and six first-order partial derivative coefficients,

$$T \longrightarrow \big(T;\ \partial_x T,\ \partial_y T,\ \partial_z T,\ \partial_Q T,\ \partial_\sigma T,\ \partial_v T\big),$$

and every arithmetic operation propagates all six coefficients by the chain rule, exactly, to machine precision. Truncation at order 1 means the cost is a small constant factor over the plain `double` build (measured $\approx 4.7\times$ for a seven-direction configuration), not the O(re-runs) of finite differencing.

Three seeding sites define *what* the derivatives mean:

- **Evaluation point** (x, y, z): each grid point's coordinates are seeded in `Grid::Calc_T`, so T at that point carries its own spatial gradient ∇T. This gradient is what converts field sensitivities into *geometry* sensitivities (Section 4).
- **Beam power** Q: seeded on the quadrature nodes of the *current (last) path segment only*, per input watt. Hence $\partial T/\partial Q$ is the exact sensitivity to *this segment's* power, with all history nodes carrying zero derivative.
- **Beam width** σ: likewise seeded on the current segment's effective lateral widths only.
- **Scan speed** v: seeded on the current segment's duration $\Delta t = L/v$. Unlike the four above, v does not appear in the integrand at all; it reparametrizes the quadrature, and so needs the separate treatment of Section 3.1.

The "current segment only" choice encodes the causality of the sequential controller: segment i's controls cannot change the already-executed history $j < i$, and correspondingly the Jacobian entries are pure per-segment control derivatives. This was verified against central finite differences (agreement to $< 1\%$, limited by FD truncation), including the negative control: an FD perturbation of a *history* segment's power does not reproduce the $\partial T/\partial Q$ column, exactly as intended.

Differentiation under the quadrature. It is worth being precise about *why* one hypercomplex pass suffices for x, y, z, Q, σ. Each of these enters the discrete field (6) *only through the summand* — never through the node set, the weights, or the upper limit of the sum. The nodes and weights are fixed by the adaptive schedule on the real-valued path history, so seeding a parameter changes only the *value* each node contributes, not *where* or *how many* nodes there are. Differentiation therefore commutes with the sum,

$$\partial_p T = \sum_n \partial_p \Big(c_n \, e^{-3\|\mathbf{x}_p - \mathbf{x}_{b,n}\|^2_{\Phi_n}} \Big) , \tag{8}$$

the *same* nodes and weights with each summand differentiated — the discrete analogue of differentiating (2) under the integral sign, carried out operation-by-operation by the OTI arithmetic. This makes the three "easy" controls concrete: Q is a linear prefactor of c_n, so $\partial_Q T = (T - T_0)/Q$; the evaluation point enters only the exponent, giving the spatial gradient $\partial_x T = \sum_n (\text{node}_n) \big(-6\,(x - x_{b,n})/\phi_{x,n}\big)$ that Section 4 turns into a geometry sensitivity; and σ enters both the exponent (through ϕ_x, ϕ_y in Φ_n) and the $1/\sqrt{\phi_x \phi_y \phi_z}$ normalization in c_n, whose two chain-rule contributions the arithmetic combines automatically. A control that lives in the integrand costs nothing but a slot.

Analytic verification. Because these five enter only the summand, their derivatives have closed forms — a reference *stronger* than finite differencing, with no step size, truncation, or subtractive cancellation. Writing $g_n = c_n \exp(-3\|\mathbf{x}_p - \mathbf{x}_{b,n}\|^2_{\Phi_n})$ for the n-th node's contribution to (6),

$$\partial_Q T = (T - T_0)/Q, \tag{9}$$

$$\partial_x T = \sum_n g_n \, \frac{-6\,(x_p - x_{b,n})}{\phi_{x,n}}, \tag{10}$$

$$\partial_\sigma T = \sum_n g_n \, \sigma \left(\frac{6\big[(x_p - x_{b,n})^2 + (y_p - y_{b,n})^2\big]}{\phi_{x,n}^2} - \frac{2}{\phi_{x,n}} \right), \tag{11}$$

with $\partial_y T, \partial_z T$ analogous to (10). Equation (10) is the dimensional spatial gradient of [1] (Appendix B, their G_x); (11) we derive here, and it is straightforward precisely because $\sigma_x = \sigma_y = \sigma$ is a single variable entering $\phi_x = \phi_y$ in *both* the exponent and the $1/\sqrt{\cdot}$ normalization (with σ_z held fixed). Evaluated against a stationary single-spot source — one segment, so the OTI columns are the *full* derivatives and the path integral collapses to a one-dimensional conduction-time integral computed independently by adaptive quadrature (`scipy`) — the OTI $\partial_x T, \partial_y T, \partial_z T, \partial_\sigma T$ agree to $\sim 10^{-6}$ relative (the solver's own forward quadrature tolerance) and $\partial_Q T$ to machine precision ($\sim 10^{-16}$ at strong signal), confirming the implementation and the derived $\partial_\sigma T$ at once. Because v instead reparametrizes the sum, it has no such closed form and is validated separately below.

3.1 Velocity: differentiating the quadrature itself

Scan speed is the exception that proves the rule, and the reason it needed work rather than a slot. At *fixed* path geometry — the raster must still cover the part — the current segment's speed v appears nowhere in the integrand; it sets the segment's *duration* $\Delta t = L/v$ and, through it, the time parametrization of the quadrature. Perturbing v moves the construction of the sum (6), not its summand, in three coupled ways:

1. the current segment's node *conduction times* and *weights* both scale with Δt — a node at path fraction ξ sits at conduction time $(1 - \xi)\Delta t$ with weight $\propto \Delta t$;

2. the snapshot is taken at scan end, so the *observation time* $t_{\text{end}} = t_{i-1} + \Delta t$ itself moves with v;
3. consequently *every history node's* conduction time $t_{\text{end}} - t'$ shifts, even though its own controls are frozen — a channel with no analogue for Q or σ.

All three funnel through the single scalar $\Delta t = L/v$. Seeding it as the sixth OTI direction, current-segment nodes take a multiplicative factor $k = \Delta t(v)/\Delta t_0$ on both conduction time and weight (real part 1), and history nodes take an additive observation shift $t_{\text{shift}} = \Delta t(v) - \Delta t_0$ on their conduction time (real part 0); one pass then delivers $\partial_v T$ exactly. Appendix A carries out the matching closed-form derivation and shows term-for-term how k and t_{shift} reproduce it. In the plain-`double` build $k = 1$ and $t_{\text{shift}} = 0$ identically, so that path stays bit-for-bit unchanged.

Validation against central finite differences: the field derivative $\partial_v T$ agrees to $\sim 0.02\%$ (median over the highest-signal grid points) on both a single-track segment and a mid-raster segment, the latter confirming the history channel (item 3) that a kinematics-only implementation would miss; the depth sensitivity agrees to $< 0.5\%$, and all signs are physical (faster $\Rightarrow$ cooler, shallower, narrower). The half-width finite difference shows larger apparent error, but only because it is read on the $50\,\mu$m lateral grid: refining that grid collapses it toward the grid-independent analytic $\partial_v w$, confirming the gap is the marching-cubes width estimate, not $\partial_v T$.

Velocity also admits an *analytic* check, and a stronger one: for a moving source the temperature is still a one-dimensional integral over the deposition time t' (with $\mathbf{x}_b(t')$ the piecewise-linear beam position), whose v-derivative — $\mathrm{d}T/\mathrm{d}v = -(L/v^2)\,\mathrm{d}T/\mathrm{d}\Delta t$ with the three channels derived in closed form in Appendix A — is evaluated independently by adaptive quadrature. On a straight multi-segment track, and at two different observation times, the OTI $\partial_v T$ matches this reference to $\sim 10^{-6}$ relative (the solver's own forward-quadrature tolerance, away from the sign changes of $\partial_v T$), together with $\partial_{x,y,z} T$ over the full path and $\partial_Q T, \partial_\sigma T$ over the last segment — i.e. every OTI direction, on the raster-relevant moving geometry, verified against a closed-form reference.

4 From field derivatives to pool-dimension derivatives

The controller regulates pool *geometry*, so the raw field derivatives must be assembled into derivatives of the measured dimensions.

Measurement. The pool boundary is the liquidus isotherm $T = T_{\text{liq}} = 1733\,\text{K}$, extracted from the snapshot grid by marching cubes. The quantities of interest are taken at the extremal vertices of that isosurface:

$$w = \tfrac{1}{2}\,(y_{\text{right}} - y_{\text{left}}), \qquad d = -z_{\text{bottom}}, \qquad a = y_{\text{right}} + y_{\text{left}} \ \text{(centre offset)}.$$

Implicit function theorem. Let $p \in \{Q, \sigma\}$ be a control and $y_e(p)$ the lateral position of an extremal edge point, defined implicitly by

$$T\big(y_e(p),\, p\big) = T_{\text{liq}}. \tag{12}$$

Differentiating,

$$\frac{\mathrm{d}y_e}{\mathrm{d}p} = -\left.\frac{\partial T/\partial p}{\partial T/\partial y}\right|_{\text{at the edge point}}, \qquad \frac{\mathrm{d}z_b}{\mathrm{d}p} = -\left.\frac{\partial T/\partial p}{\partial T/\partial z}\right|_{\text{at the bottom point}}. \tag{13}$$

At the *widest* points of the isotherm the surface is tangent to the x–z plane, so the isotherm normal is aligned with $\pm y$ and the one-dimensional relation (13) is the exact first-order edge sensitivity (similarly $\pm z$ at the deepest point). Both numerator and denominator are OTI coefficients interpolated (trilinearly) at the fractional-index marching-cubes vertices; no additional simulation is involved. Assembly is then linear:

$$\frac{\mathrm{d}w}{\mathrm{d}p} = \frac{1}{2}\left(\frac{\mathrm{d}y_r}{\mathrm{d}p} - \frac{\mathrm{d}y_l}{\mathrm{d}p}\right), \qquad \frac{\mathrm{d}d}{\mathrm{d}p} = -\frac{\mathrm{d}z_b}{\mathrm{d}p}, \qquad \frac{\mathrm{d}a}{\mathrm{d}p} = \frac{\mathrm{d}y_r}{\mathrm{d}p} + \frac{\mathrm{d}y_l}{\mathrm{d}p}. \tag{14}$$

These four numbers ($\mathrm{d}w/\mathrm{d}Q$, $\mathrm{d}w/\mathrm{d}\sigma$, $\mathrm{d}d/\mathrm{d}Q$, $\mathrm{d}d/\mathrm{d}\sigma$) are the exact Jacobian of the control problem, obtained from *one* simulation.

Generic form. The implementation (`ExtractIsoSupportSensitivities`) does not hardcode the axes or the control set. For an arbitrary unit direction $\mathbf{e}$, the isosurface point extremal in $\mathbf{e}$ (the *support point* $\mathbf{x}^*(\mathbf{e})$) has $\nabla T \parallel \mathbf{e}$, and the support value $h(\mathbf{e}) = \mathbf{e} \cdot \mathbf{x}^*$ obeys

$$\frac{\mathrm{d}h(\mathbf{e})}{\mathrm{d}p} = -\left.\frac{\partial T/\partial p}{\nabla T \cdot \mathbf{e}}\right|_{\mathbf{x}^*(\mathbf{e})}, \tag{15}$$

with the full gradient $\nabla T = (\partial_x T, \partial_y T, \partial_z T)$ taken from the snapshot's OTI columns and the controls p *discovered* as every remaining `dT_d`$\langle$*name*$\rangle$ column. The melt-pool QoIs are then just linear combinations, $w = \frac{1}{2}\big(h(+\hat{y}) + h(-\hat{y})\big)$, $d = h(-\hat{z})$, $a = h(+\hat{y}) - h(-\hat{y})$, and likewise for their derivatives. A 3DThesis build with additional seeded design variables (velocity, material properties, ...) is picked up by this layer with no code change.

5 The sequential per-segment Newton controller

Segments are optimized in path order (as the real process would execute them). For segment i with controls $\mathbf{u} = (P_{\mathrm{mod}}, \sigma)$, define the residual

$$\mathbf{r}(\mathbf{u}) = \begin{pmatrix} w(\mathbf{u}) - w^* \\ d(\mathbf{u}) - d^* \end{pmatrix}, \qquad J(\mathbf{u}) = \begin{pmatrix} \partial w/\partial P_{\mathrm{mod}} & \partial w/\partial \sigma \\ \partial d/\partial P_{\mathrm{mod}} & \partial d/\partial \sigma \end{pmatrix}, \tag{16}$$

where each evaluation of $\mathbf{r}$ *and* J is one OTI simulation of the path truncated at segment i, with all history segments $j < i$ replaying their already-optimized $(P_{\mathrm{mod},j}, \sigma_j)$ (sequential causality; the per-segment width column carries $\sigma_j/\sigma_{\mathrm{base}}$).

A plain Newton step $\Delta\mathbf{u} = -J^{-1}\mathbf{r}$ is fragile here (controls of different scales, near-singular J when inherited heat dominates, bound slamming), so the implemented step is a scaled, regularized Gauss–Newton with safeguards:

1. **Normalize**: residuals by the targets, $\tilde{\mathbf{r}} = \mathbf{r} \oslash (w^*, d^*)$; controls by characteristic scales $(1,\ 10\,\mu\mathrm{m})$.

2. **Regularized step**: solve $(\tilde{J}^{\mathsf{T}}\tilde{J} + \lambda I)\,\Delta\mathbf{z} = -\tilde{J}^{\mathsf{T}}\tilde{\mathbf{r}}$ with Tikhonov parameter $\lambda = 2.5 \times 10^{-2}$ in scaled coordinates.

3. **Trust region**: clip $\Delta\mathbf{u}$ componentwise to $(0.6,\ 10\,\mu\mathrm{m})$ per iteration.

4. **Merit backtracking**: accept the trial only if the merit $\phi = \frac{1}{2}\|\tilde{\mathbf{r}}\|^2$ decreases; otherwise halve the step ($\alpha = 1, \frac{1}{2}, \ldots, \frac{1}{16}$). Each trial is itself an OTI simulation and is cached/reused as the next iterate's evaluation when accepted.

5. **Bounds**: $P_{\text{mod}} \in [0.05, 3]$, $\sigma \in [5, 60]\,\mu$m. If no backtracked step improves the merit, the segment returns its *best evaluated* state.

6. **Warm starts**: the initial guess for a segment is the converged control of the *same within-line position one line below* — the raster is line-periodic, so this is a far better prior than the previous segment (a line-end control is a poor guess for the next line's fresh-ground start).

Convergence is declared at $\max_k |r_k|/\text{target}_k < 1\%$. In the demo this takes 1–4 accepted iterations per segment ($\approx$2.3 on average) and $\approx$3.5 simulations per segment including backtracking trials — 49 simulations for all 14 controlled segments, about 65 s wall clock in total (single process; each local-domain OTI solve is $\approx$1.3 s).

6 Feasibility: three things the controller cannot fix

Three physical constraints shape the problem setup; getting these wrong makes the target unreachable no matter how good the optimizer is.

Turnaround dwell. At $H = 0.15$ mm the return pass starts next to the previous track's still-hot end. With an instantaneous turnaround the new pool *merges with still-liquid residue*; the measured width then contains inherited melt that no control value can remove, and a fixed width target is infeasible at track starts. The demo therefore inserts the *minimal* dwell such that the whole field has solidified when each new track starts, found by bisection on $\max_{\mathbf{x}} T < T_{\text{liq}}$ at the end of every dwell (at worst-case 60 W power): $t_{\text{dwell}} = 1.234$ ms.

Target selection. A uniformity target must be a *sustainable fixed point* of the controlled process: reachable from a fresh track start at moderate power *and* holdable mid-line within the control bounds. The demo uses the developed single-track pool (the last segment of line 0, $w^* = d^* = 97.5\,\mu$m). Targeting the heat-soaked centre-of-square pool instead (a snapshot of the *defect*, not the goal) forces a $\sim$175 W spike at every line start whose surplus heat the following segments cannot control, and the schedule degenerates into a spike-and-coast limit cycle.

Control-segment length. The pool has thermal inertia: after a segment's beam moves on, its superheated core keeps melting outward for roughly $\tau \approx w^2/4\alpha \approx 0.4$ ms. Control segments shorter than this (0.25 mm = 0.36 ms at 0.7 m/s) produce a period-2 "ping-pong" — alternate segments hit the target and the ones in between catch their predecessor's afterglow. With 0.5 mm segments ($\approx$2 relaxation times) each segment reaches its own quasi-steady pool before it is measured, and the cycling vanishes.

7 Results

The optimized schedule is smooth and interior: line 0 runs at the nominal (60 W, $\sigma = 10\,\mu$m); once neighbour preheat exists the power settles to $\approx$41–48 W with σ near 9.6–10 μm (Figure 1).

	width CV (%)	depth CV (%)	w range (μm)	d range (μm)
unoptimized (constant 60 W)	9.72	7.14	38.3	26.7
optimized (P_{mod}, σ per segment)	**0.71**	**0.59**	**2.3**	**1.9**

Table 1: Melt-pool uniformity over the raster, measured on the 14 controlled segments. 49 OTI simulations, ≈65 s wall clock single-process.

MPI operation was verified separately: with the domain-decomposed 3DThesis build the same optimization reproduces the serial schedule bit-for-bit (rank snapshot slices must be re-sorted into grid order before measurement). At this problem size MPI is *not* faster (≈1 s solves; launch and merge overhead dominates); it pays off on larger domains, where the same build scales ≈ 2× at 4 ranks.

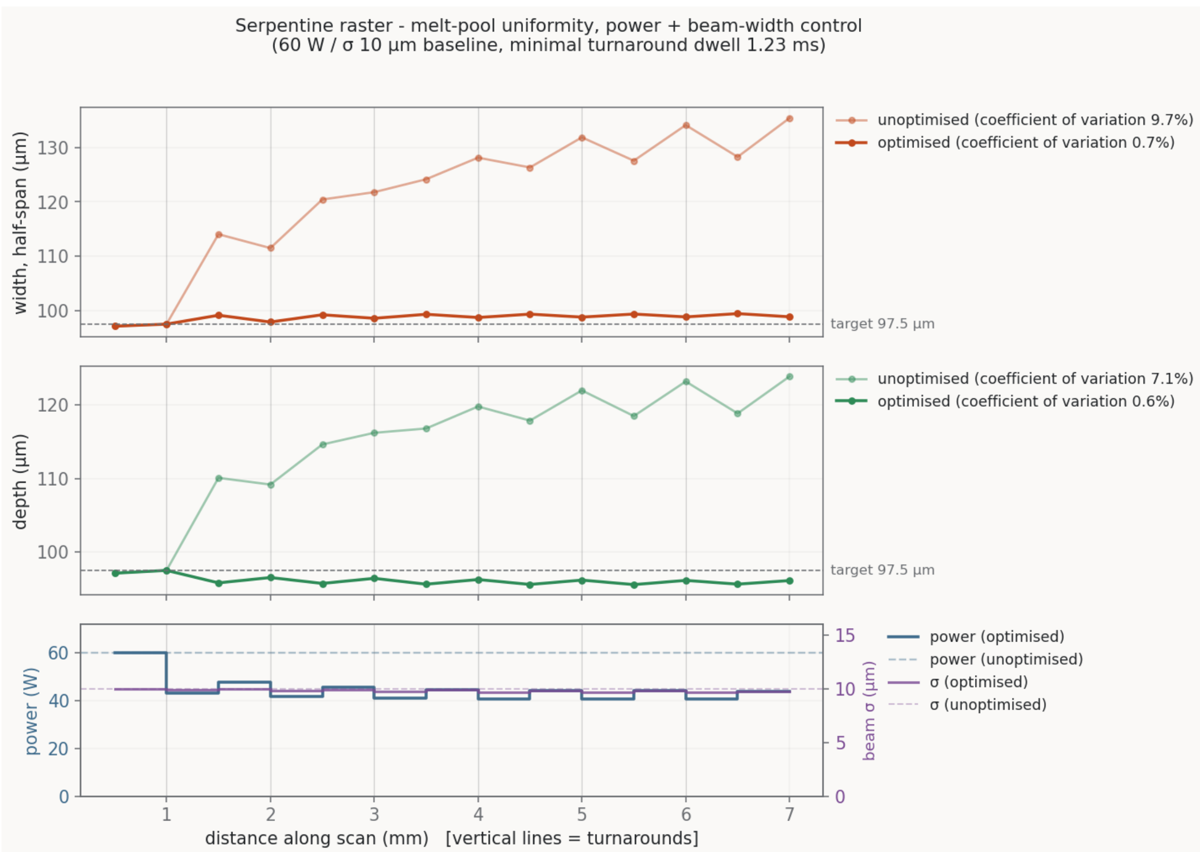

Figure 1: Width and depth along the scan path, unoptimized vs. optimized, with the optimized power and beam-width schedules (bottom panels).

8 Scaling up: a 10 mm square at published EBM parameters

The raster demo above lives in a compact-pool regime: a ∼100 μm hemispherical pool, small against both the line length and the hatch. To prepare the optimization of Stump and Plotkowski's

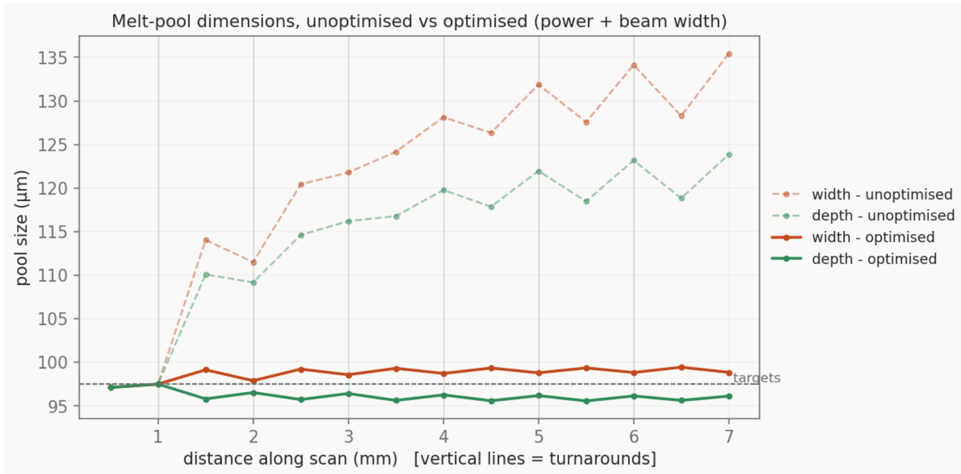

Figure 2: All four dimension traces on one axis: the optimized width and depth collapse onto the target while the unoptimized traces drift and sawtooth.

triangle experiment (their §3.5), the study was repeated on a 10×10 mm square at that experiment's physics — IN718 at $T_0 = 1273\,\mathrm{K}$, $V = 3\,\mathrm{m/s}$, $\sigma_{xy} = 200\,\mu\mathrm{m}$, $\sigma_z = 10\,\mu\mathrm{m}$, $Q = 150\,\mathrm{W}$ absorbed, $T_{\mathrm{liq}} = 1610\,\mathrm{K}$, hatch $H = 0.1$ mm (`melt_pool_demos/square`; the same scripts drive the triangle geometry). The square keeps the line length constant, so the process reaches a statistically steady state that the triangle's shrinking lines never do.

Regime change. A single-track calibration run sets the scales: the developed pool is 2.05 mm long (1.95 mm trailing), 0.30 mm wide, and 60 μm deep, with a 0.55 ms post-beam solidification tail. Because the beam ($\sigma_{xy} = 200\,\mu$m) is wider than two hatch spacings, the liquid spans many tracks laterally — a melt *front* rather than discrete tracks. Everything resizes accordingly: control segments of 5 mm (two per line, ≥ 2 pool-relaxation lengths), a measurement box extending 7 mm behind the segment end, and the target taken again from the developed single track of line 0 ($w^* = 163.8\,\mu$m half-span, $d^* = 63.3\,\mu$m on the snapshot grid).

Two dwell policies. The published experiment scans continuously, so zero dwell is studied alongside the raster demo's minimal worst-case dwell. The minimal dwell — bisected on the RDF event list with the criterion *no liquid cell anywhere at any line-start instant* — is 0.906 ms/turn, binding at a mid-square turn (the steady regime, unlike the small raster where accumulation made the last turn binding), and adds 26% to the build time. At zero dwell the field *never* solidifies between lines: every line start inherits $\approx 0.06\,\mathrm{mm}^3$ of still-liquid residue, about three single-track pools' worth. The target is deliberately kept identical (the well-formed single-track pool, never the heat-soaked one), and any resulting infeasibility is left to appear honestly as bound-pinned segments rather than hidden by re-measuring. In the continuous path the 0.1 mm serpentine hops stay powered; they are excluded from the control set and execute the controls of the segment they follow.

How the results are judged. The controller regulates snapshot dimensions at the 202 segment ends, but the part does not care about the control grid, so results are reported on *whole-build* statistics from full solidification-tracked replays of the optimized schedules: the mean and standard deviation of the *instantaneous* pool depth, length, and volume over every beam-on instant (50 μs sampling from the RDF event list). Beam-off dwell windows are excised — a pool solidifying during a designed dwell is the end of that pool's life, not process variation. Fluctuations are quoted in absolute units rather than as CVs: the optimizer shrinks the mean pool together with its spread, so relative measures hide the reduction, and part defects (remelt geometry, microstructure gradients) live in absolute units. The companion spatial view is the *fusion-depth map*: the deepest melt each interior (x, y) column attained over the entire build — a max-attained quantity that cooling never touches.

beam-on mean $\pm\sigma$	depth (μm)	length (mm)	lateral extent (mm)	volume (mm^3)	fusion depth (μm)
zero dwell, unoptimized	106 ± 12.9	4.24 ± 1.23	0.34 ± 0.035	0.077 ± 0.024	106 ± 10.7
zero dwell, optimized	$\mathbf{69 \pm 7.2}$	$\mathbf{2.22 \pm 0.47}$	0.31 ± 0.016	$\mathbf{0.020 \pm 0.005}$	67 ± 6.7
minimal dwell, unoptimized	94 ± 15.4	3.40 ± 1.36	0.31 ± 0.034	0.054 ± 0.021	95 ± 8.5
minimal dwell, optimized	$\mathbf{64 \pm 8.7}$	$\mathbf{1.99 \pm 0.55}$	0.30 ± 0.026	$\mathbf{0.017 \pm 0.004}$	64 ± 4.4

Table 2: The 10 mm square at the published parameters: whole-build pool statistics (beam-on instants of the tracked replays) and the fusion-depth map over interior columns. "Lateral extent" is the y-span of all liquid — at zero dwell this includes the neighbouring track's still-molten band — not the beam-frame track half-width the controller regulates. Each optimization used ≈220 OTI simulations (1.1 per segment — line-periodic warm starts let most segments converge on their first evaluation) in ≈25 min single-process. Depth statistics carry the 5 μm z-grid quantization of these replays.

What the optimizer achieves. The absolute pool fluctuation drops $\approx 2\times$ in depth, $2.5\times$ in length, and $4.5\times$ in volume (Table 2, Figure 3), while the mean is simultaneously pinned to the requested setpoint (67/64 μm attained vs. 63.3 requested; the constant-power baselines run 50–65% too deep). The residual fluctuation is dominated by a once-per-line startup transient — the pool briefly overshoots to $\approx 85\,\mu$m over the first 1–2 mm of each line, where the fresh track crosses its neighbour's still-hot end — a sub-segment feature invisible to a controller whose actions are constant per 5 mm segment and whose residual is evaluated after the transient has relaxed. Regulating it needs sub-segment authority: a within-segment peak term in the residual, graded segment lengths near the turns, or the velocity and dwell controls of Section 8. As controller diagnostics (did the Newton converge — not part quality): the segment-end width/depth CVs fall from 3.5/7.2% to 0.12/0.21% (minimal dwell) and 4.2/10.3% to 0.19/0.36% (zero dwell), with **no segment pinned at a control bound** in either policy — the single-track target is feasible everywhere on this geometry.

The price of zero dwell. A genuine trade-off rather than a free lunch in either direction: the dwell buys a modestly tighter build at part level (fusion-depth σ 4.4 vs. 6.7 μm; beam-on volume σ 0.0044 vs. 0.0053 mm^3) at the cost of +26% build time (428 vs. 341 ms). Both optimized schedules are smooth, interior, and line-periodic; the zero-dwell schedule runs ≈5 W cooler because inherited background heat performs part of the melting, and it leaves $3.5\times$ less residual liquid at line starts than the constant-power baseline (0.017 vs. 0.059 mm^3 mean) — holding the target forces a cooler schedule, which attacks the merged-pool defect indirectly.

The triangle. The same machinery (environment-selected geometry) runs the triangle raster: 131 controlled segments over 87 shrinking lines, same target. Its minimal worst-case dwell is 1.672 ms/turn with the *binding turn near the apex* (#79) — that global dwell takes the 149 ms continuous path to 290 ms (+94% build time, versus +26% for the square), which is the quantitative case for the adaptive per-turn dwell below. At nominal power the shrinking lines drive the pool from the 164 μm single-track half-width at the base to an 875 μm lake near the apex; on whole-build statistics the optimizer cuts the pool-volume fluctuation 7× at zero dwell ($0.113 \pm 0.048 \to 0.022 \pm 0.007\,\text{mm}^3$; beam-on depth $116 \pm 21 \to 68 \pm 9\,\mu$m, fusion-depth map $111 \pm 25 \to 65 \pm 12\,\mu$m) and 4× with the dwell. The residual variation concentrates entirely at the apex, where the bound-pinning diagnostic localizes the physics: segment-end CVs are square-quality over the body (w/d 0.2–0.3%, lines 0–69) but degrade over the last ∼17 lines, and the two policies fail in *opposite* directions. With the dwell (solidified starts), the 0.07 mm tip line pins at **maximum power** (450 W) yet remains 21 μm short of target depth — 23 μs of beam time cannot develop a pool, and the physical fix is *slowing down.* At zero dwell, lines 80–83 pin σ at its floor and lines 84–86 pin at **minimum power** (8 W) with the pool still at or above target width — pure inherited melt, and the fix is *waiting* (dwell) or again slowing down. Between them, the apex brackets exactly the two control degrees of freedom developed next.

9 Limitations and next steps

- **Velocity as a control.** Scan speed is fixed at the raster speed. Velocity is the missing degree of freedom whenever σ pins at its lower bound and the depth-to-width ratio cannot be restored. The exact sensitivity $\partial_v T$ is now implemented and validated as the sixth OTI direction (Section 3.1); what remains is wiring it into the per-segment solve. Because v and Q move the pool in nearly opposite directions in (w, d) space — both dominated by depth — the natural formulation is time-optimal (maximize v subject to $(w, d) = (w^*, d^*)$, with Q, σ as feasibility controls) rather than adding v as a third co-regulated knob; power saturating at its upper bound is the expected binding constraint on the achievable speed-up.

- **First-order control model.** The 2×2 Jacobian is exact but local; strong nonlinearity near bounds is handled by damping rather than curvature. Second-order OTI jets (`otinum<M,2>`) would provide exact Hessians at the same single-run structure if a trust-region Newton on the merit function ever becomes necessary.

- **Per-segment greedy horizon.** Each segment is solved given a frozen history; there is no look-ahead. A receding-horizon variant would re-use the same OTI Jacobians.

- **Adaptive dwell.** The global worst-case dwell is wasteful: it is sized by the single binding turn (+26% build time on the square; nearly 2× on the triangle, where the apex binds). Since a dwell simply ages all previously deposited heat, $\partial T/\partial t_{\text{dwell}}$ is seedable exactly like the other OTI directions, and a per-turn Newton solve on the solidification criterion would replace the current bisection-by-resimulation with one or two simulations per turn — the same trick this whole document is about, applied to the feasibility constraint instead of the target.

A Derivation of the velocity sensitivity

The spatial and power derivatives (§3) follow by differentiating the integrand of the moving-source solution (2) — the paper's Eq. (2) — over a *fixed* integration domain, exactly as its Appendix B obtains the thermal gradients $G_{x,y,z}$. Velocity is different: it appears nowhere in the integrand, only in the *parametrization* of the integral, so the domain itself moves. We carry the derivation out here, both to justify the closed-form reference of §3.1 and to show, in the open, what the OTI arithmetic does silently.

Write the scan-end field as an integral over deposition time t',

$$T - T_0 = A \int_0^{t_{\text{end}}} K(t')\,\mathrm{d}t', \quad K = \frac{\exp\big(-3\sum_i r_i^2/\phi_i\big)}{\sqrt{\phi_x \phi_y \phi_z}}, \quad \phi_i = 12\alpha\,(t_{\text{end}} - t') + \sigma_i^2, \tag{17}$$

with $A = 2\eta Q/\rho c\,(\pi/3)^{3/2}$, $r_i(t') = x_{p,i} - x_{b,i}(t')$ the offset from the moving beam, and $t_{\text{end}} - t'$ the conduction time. Perturb the speed v of the *last* segment only, at fixed path geometry; both effects funnel through its duration $\Delta t = L/v$ (L its fixed length):

1. the observation is the scan end, so $t_{\text{end}} = t_{N-1} + \Delta t$, with t_{N-1} (the instant the segment begins) frozen by the history;
2. within the segment the beam is at fixed fraction $\xi \in [0,1]$ of its *path*, $x_b = x_{N-1} + \xi L$, but *deposited* at $t' = t_{N-1} + \xi \Delta t$ — the deposition times stretch with Δt.

Point (2) is why one may not differentiate (17) under a fixed t': the map from physical position to deposition time depends on v (naive Leibniz at fixed t' omits $\partial x_b/\partial \Delta t$, since at fixed absolute time a slower beam is less far along). We split at t_{N-1} and treat each piece on its natural, v-independent domain. Using $\partial K/\partial \phi_i = K\,(3r_i^2/\phi_i^2 - 1/2\phi_i)$, abbreviate

$$S(t') \equiv \sum_i \left(\frac{3r_i^2}{\phi_i^2} - \frac{1}{2\phi_i} \right), \qquad \text{so} \qquad \frac{\partial K}{\partial \Delta t} = 12\alpha\, w\, K\, S \tag{18}$$

whenever a change in Δt shifts every ϕ_i by the common amount $12\alpha\, w\, \mathrm{d}\Delta t$.

History, $0 \le t' \le t_{N-1}$. The limits and deposition times are fixed; Δt enters only through t_{end} in $\phi_i = 12\alpha(t_{N-1} + \Delta t - t') + \sigma_i^2$, so $w = 1$ and

$$\frac{\mathrm{d}}{\mathrm{d}\Delta t} \int_0^{t_{N-1}} K\,\mathrm{d}t' = 12\alpha \int_0^{t_{N-1}} K\,S\,\mathrm{d}t' \;\equiv\; H. \tag{19}$$

Every history node's Gaussian broadens because the field is observed a little later — the t_{shift} channel.

Last segment, $t_{N-1} \le t' \le t_{\text{end}}$. Change to the fixed variable ξ: $t' = t_{N-1} + \xi \Delta t$, $\mathrm{d}t' = \Delta t\,\mathrm{d}\xi$, whence $t_{\text{end}} - t' = (1-\xi)\Delta t$ and $\phi_i = 12\alpha(1-\xi)\Delta t + \sigma_i^2$, with v-independent limits. Differentiating the fixed-limit form splits into a Jacobian and an integrand term,

$$\frac{\mathrm{d}}{\mathrm{d}\Delta t}\Big[\Delta t \int_0^1 \tilde{K}\,\mathrm{d}\xi\Big] = \underbrace{\int_0^1 \tilde{K}\,\mathrm{d}\xi}_{W} \;+\; \underbrace{\Delta t \int_0^1 \frac{\partial \tilde{K}}{\partial \Delta t}\,\mathrm{d}\xi}_{C}, \tag{20}$$

where W is the weight growing because the segment lasts longer, and C uses $w = (1-\xi)$ in (18) (a node early in the segment ages more when it is stretched; the node at the observation instant, $\xi = 1$, not at all). Restored to t',

$$W = \frac{1}{\Delta t} \int_{t_{N-1}}^{t_{\mathrm{end}}} K \,\mathrm{d}t', \tag{21}$$

$$C = \frac{12\alpha}{\Delta t} \int_{t_{N-1}}^{t_{\mathrm{end}}} (t_{\mathrm{end}} - t')\, K\, S \,\mathrm{d}t'. \tag{22}$$

Result. With $\mathrm{d}\Delta t/\mathrm{d}v = -L/v^2$,

$$\frac{\mathrm{d}T}{\mathrm{d}v} = -\frac{L}{v^2}\, A\,(H + W + C), \tag{23}$$

the three integrals evaluated by the closed-form reference of §3.1: W and C the last-segment weight and conduction-time channels, H the history channel.

Forward-mode AD never forms (19)–(22). It seeds the single scalar $\Delta t = L/v$ with $\partial \Delta t/\partial v = -L/v^2$, and facts (1)–(2) enter as the two node dressings of §3.1: current-segment node conduction times and weights carry $k = \Delta t/\Delta t_0$, history-node conduction times carry $t_{\mathrm{shift}} = \Delta t - \Delta t_0$. Propagating those through K node-by-node and summing — one pass — reproduces (23) term for term: the derivative of k feeds W (its weight part) and C (its conduction part), and that of t_{shift} feeds H. The derivation above is only the closed-form view of the chain rule the hypercomplex numbers carry automatically.

References

[1] B. Stump and A. Plotkowski, "An adaptive integration scheme for heat conduction in additive manufacturing," *Applied Mathematical Modelling* **75** (2019) 787–805. doi:10.1016/j.apm.2019.07.008.

[2] N. T. Nguyen, A. Ohta, K. Matsuoka, N. Suzuki, and Y. Maeda, "Analytical solutions for transient temperature of semi-infinite body subjected to 3-D moving heat sources," *Welding Journal* **78** (1999) 265s–274s.

[3] `cpp_oti_lib`: header-only static order-truncated imaginary (OTI) numbers for C++ (`oti::otinum<M,N,T>`), used here as the 3DThesis scalar type with $M = 5$ directions at order $N = 1$.

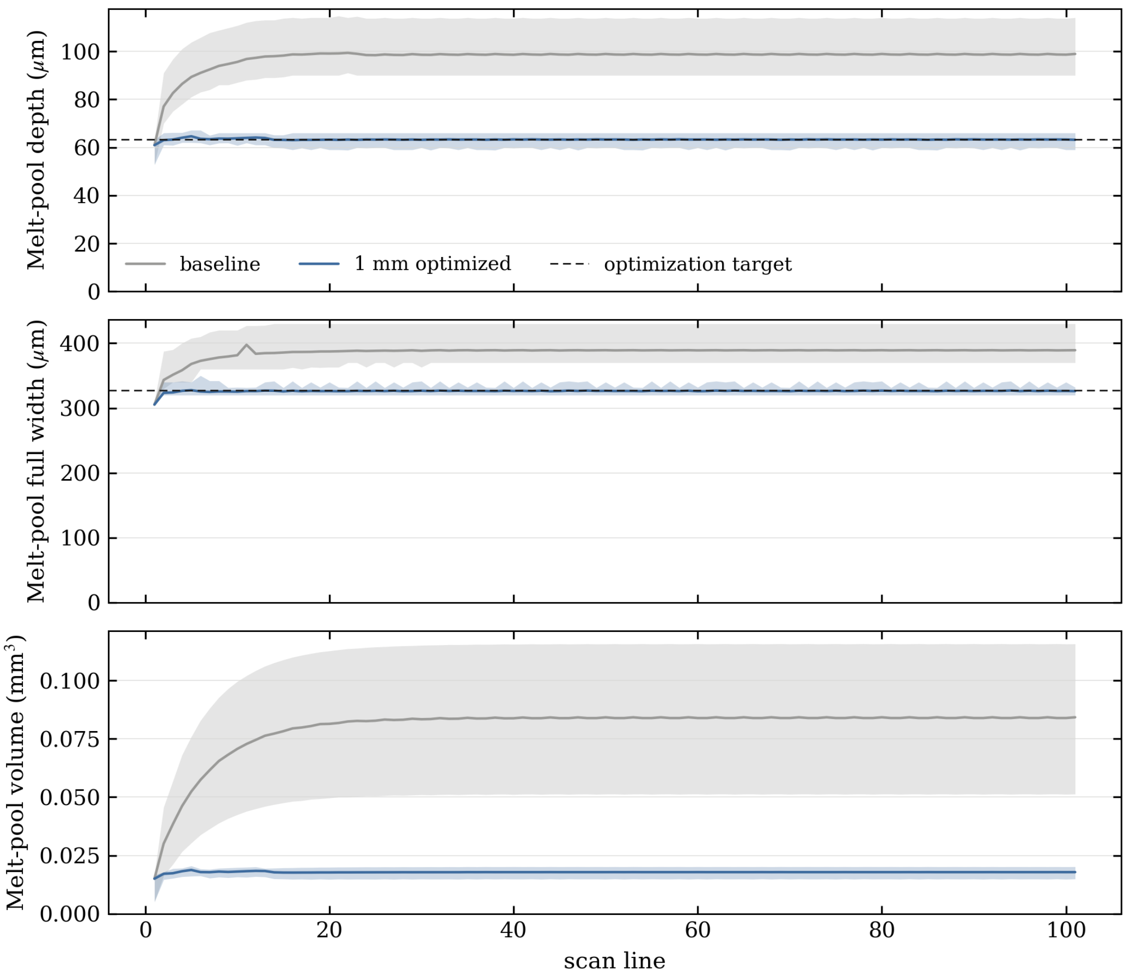

Figure 3: Per-line pool statistics of the zero-dwell square (RDF event list, constant-power baseline vs. the replayed optimized schedule): for each scan line, the mean (solid) and the 5th–95th percentile band (shaded) of the instantaneous beam-on depth, volume, and length, annotated per panel with the whole-build beam-on mean $\pm\,\sigma$ of Table 2. The bottom strip resolves three mid-build lines in real time (marked in the depth panel).

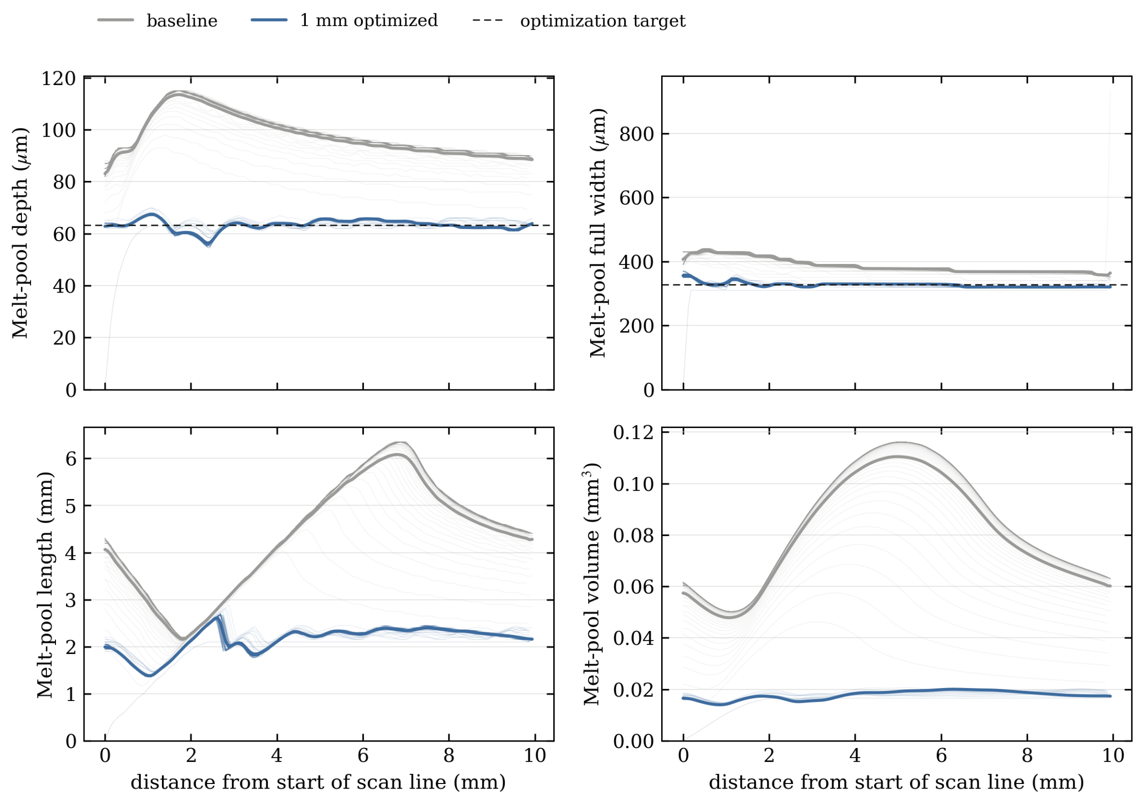

Figure 4: The same zero-dwell data folded per line: all 101 lines overlaid against position along the line (light curves), with the per-case mean (bold). After the ∼15-line startup the bundles are nearly one-dimensional — line-to-line variation is essentially eliminated by the statistically steady process and, for the optimized schedule, the remaining imperfection is the repeatable within-line structure: the inherited pool at position 0, its die-off near 1 mm, and the development hump at 1.5–3.5 mm that the per-segment controller cannot resolve.

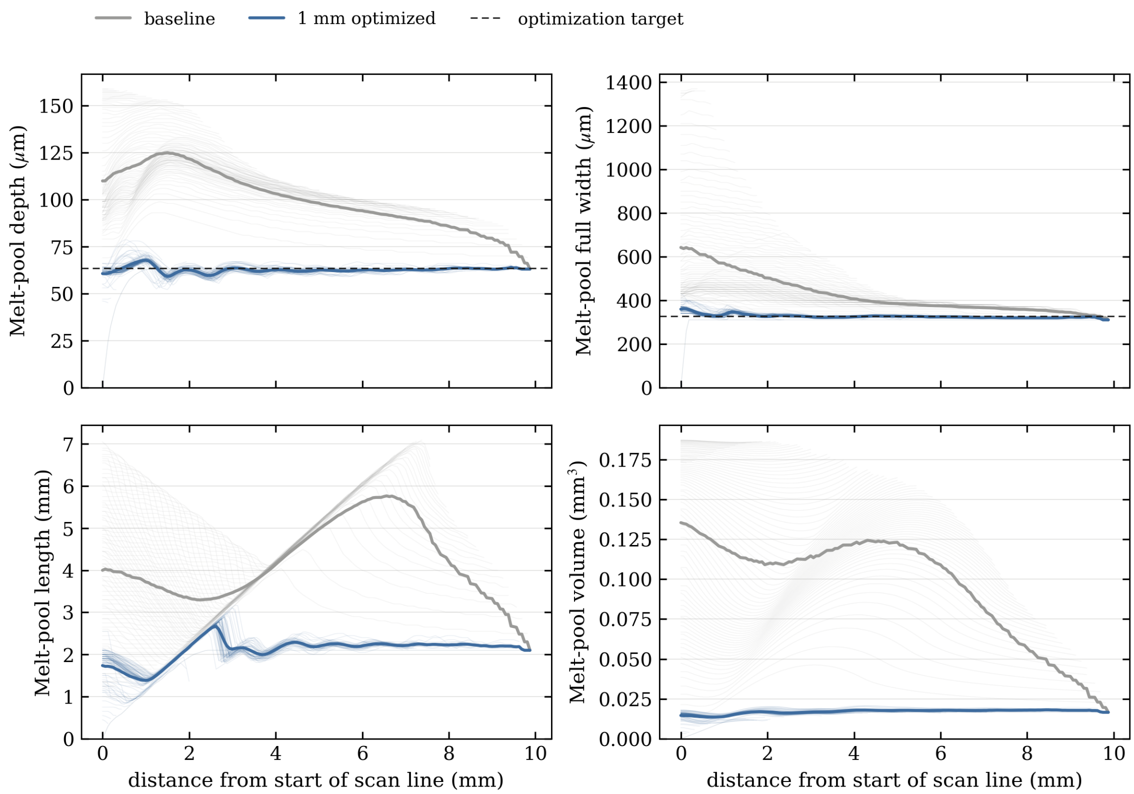

Figure 5: The zero-dwell triangle folded per line (all 87 lines): the unoptimized fan — every line a different length, the pool deepening and widening toward the apex — collapses onto the target band under the optimized schedule, with the apex lines carrying the residual variation that power and beam width cannot remove.

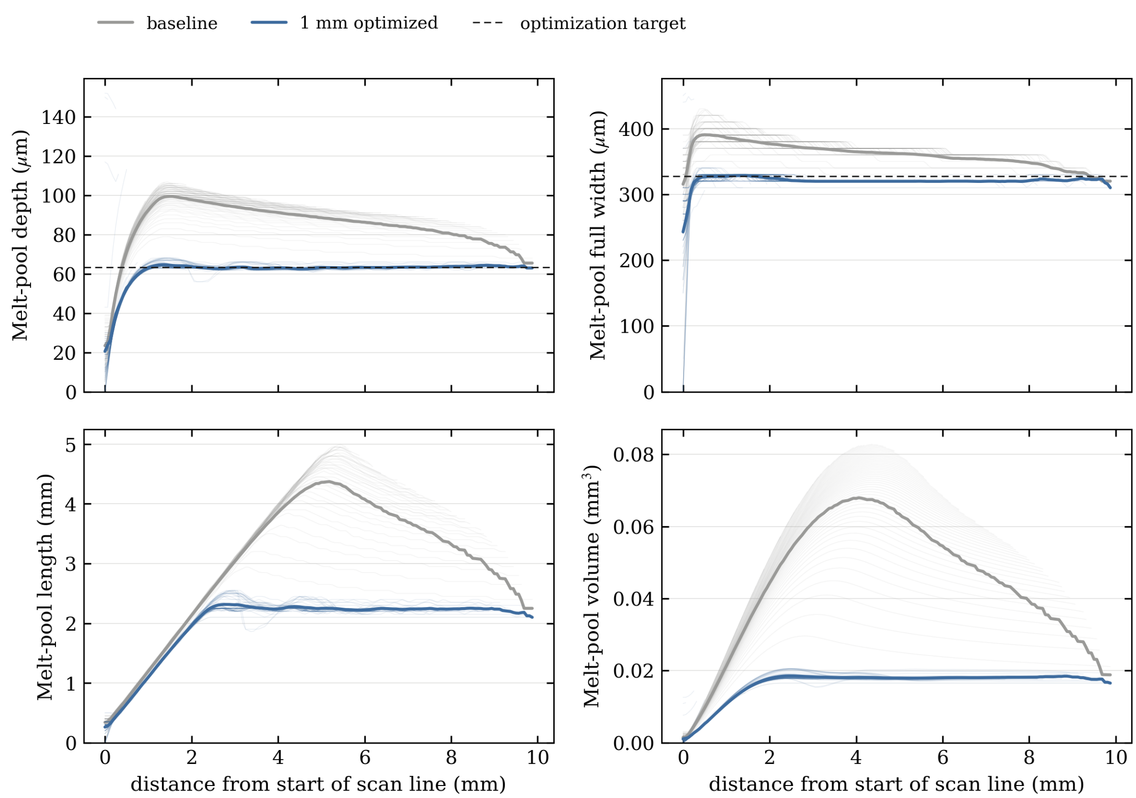

Figure 6: The minimal-dwell triangle folded per line: with solidified starts every pool is reborn from cold ground, so the profiles carry the development ramp once per line; the optimized bundle rides the target until the shortest apex lines, whose pools cannot develop within the available beam time even at the 450 W power ceiling.